

Linear Modeling Analysis of Cattle Performance in an East Java Feedlot

1 Executive Summary

This document is about analysing the performance of Bulls in Jombang, East Java, Indonesia. Multiple types of models, SIMEX, OLS, PanelOLS, and mixed effects. Multiple different model specifications were tried, specifically ‘naive’ specifications that used common feedlot logic and metrics, and models based on the formulas found in *Nutrient Requirements Of Beef Cattle, 8th edition, by the National Academy of Science*.

The most effective model is a Panel model with:

- R-squared (Between): 0.8510
- R-squared (Within): 0.6095
- R-squared (Overall): 0.8762

Findings: Decrease in ADG is mostly a component of metabolic weight gain times the percentage increase since the start weight. Different breeds (limousine and Simental) differ drastically in performance. Genetic uncertainty/diversity plays a big role in overall cattle performance; cattle can differ $\pm 50\%$ from the average, and the overall impact of the cattle’s innate potential was around 26.6%. Elephant grass seems to outperform all other feed types, including the slobber mix.

2 Introduction and Problem Statement

2.1 Background

The cattle industry in Jombang, East Java represents a significant sector of Indonesia’s agricultural economy. Optimizing cattle performance in feedlot settings has direct implications for profitability, resource efficiency, and sustainable livestock production. This study focuses on cattle in the Jombang region of East Java, with particular emphasis on the Limousin breed, with some observation on the Simental breed, which are common in Indonesian feedlot operations.

2.2 Research Objectives

- Develop predictive models for Average Daily Gain (ADG) in feedlot cattle
- Identify key factors influencing cattle growth performance
- Quantify the relationships between feed intake, weight, breed, and growth
- Provide actionable insights for optimizing feedlot management

2.3 Data Overview

The dataset comprises longitudinal measurements of cattle in feedlot settings, including:

- Weight measurements over the span of 10 months, at 10-day intervals.
- Feed intake records (including silage, grass, and concentrate components)
- Breed information (primarily Simmental and Limousin)
- Derived metabolic weight metrics
- 400+ observations (depending on how the data is processed) across 28 individual cattle
- Observation periods ranging from 10 to 20 days per animal, but are almost exclusively 10.

3 Methodology

3.1 Data Preprocessing

The data processing gave a unique set of problems, because the cattle were measured at a 10 day interval, and cattle can oscillate around 20 kg of body weight on a given day, depending on feed and water intake. The average ADG over the whole period was around 1 kg/day, so on a 10-day interval, that means 10 kg. This means we are trying to measure a 10 kg signal, on ± 10 kg noise.

Taking longer intervals, like using only ever uneven measurements, did not work because some predictors (like total dry matter intake) would contain future information. The weighing process does happen every time at the same time before the cattle have received their feed.

3.1.1 Kalman Filter

This is why a Kalman Filter is applied to the weight measurements. After applying the Kalman filter it identifies:

Measurement noise variance (σ_{obs}^2):

- Mean: 60.7233
- Std: 44.0470
- Min: 0.0000
- Max: 163.4434

Process noise variance (σ_{state}^2):

- Mean: 21.5999
- Std: 49.7599
- Min: 0.0000
- Max: 225.0329

Average noise: ± 7.8 kg

Average gain: 4.6 kg

With AIC values ranging from 9.04 to 246.15, depending on the animal being measured. The average noise aligns with the theoretical ± 10 kg.

3.1.2 Outlier Cutting

The model suffers from heteroscedastic errors when ADG is very high (> 2.0) or very low (< 0). These can still be measurement errors even after Kalman filtering. The low-end effects may be linked to medical state; medical records were inconsistent. The high-end likely reflects omitted-variable bias. Removing the tails increases R and P-values significantly, except for BEV features due to data loss.

3.2 3.2 Model Development

We progressively developed multiple modeling approaches:

3.2.1 3.2.1 Baseline OLS Models

In this case, OLS cannot be used because the multiple measurements per cow violate the independence of observations assumption. There is an OLS class in the code, and it can be used to play with.

3.2.2 3.2.2 Mixed Linear Models

This model fails after the Kalman filter is applied because the variance between measurements of some of the cows is too small.

3.2.3 3.2.3 Fixed Effects Panel Data Models

This is the approach that was taken, the Random intercept is calculated in a different way that does not depend on in-group sample variation, so this works. The baseline model is specified as:

$$\text{ADG}_{it} = \alpha_i + \beta_1(\text{weight})_{it} + \beta_2(\text{total_dmi}) + \beta_3(\text{total_dmi_log}) + \beta_4(\text{FeedRatio}) + \beta_4(\text{DayDifference}) + \varepsilon_{it}$$

Where:

- `adg` = kilos gained per day
- `total dmi` = Total Dry Matter Intake over the whole period
- `Total DMI log` = the log of Total dry matter intake over the whole period
- `Feed Ratio` = Dry matter intake of greens / Dry matter intake of Slobber over the whole period
- `Day Difference` = the number of days until the next measurement.

3.2.4 3.2.4 Fixed Effects Panel Data Final Models

ADG Model

$$\text{ADG}_{it} = \alpha_i + \beta_1(\text{hasBEF_dmi_dt})_{it} + \beta_2(\text{mw_dmi_dt_ratio})_{it} + \beta_3(\text{tdn_silage_dt})_{it} + \beta_4(\text{tdn_rumput_dt})_{it} + \beta_5(\text{tdn_slobber_dt})_{it} + \beta_6(\text{day_diff_dmi})_{it} + \varepsilon_{it}$$

Where:

- `hasBEF_dmi_dt` = Indicator for beef-focused diet type, normalized by days and metabolic weight
- `mw_dmi_dt_ratio` = Metabolic weight to dry matter intake ratio over time
- `tdn_silage_dt` = Total Digestible Nutrients from silage per day
- `tdn_rumput_dt` = Total Digestible Nutrients from grass (rumput) per day
- `tdn_slobber_dt` = Total Digestible Nutrients from concentrate (slobber) per day
- `day_diff_dmi` = Time interval effect on dry matter intake

FCR Model

$$\text{FCR}_{it} = \alpha_i + \beta_1(\text{hasBEF})_{it} + \beta_2(\text{mw_ratio})_{it} + \beta_3(\text{tdn_silage_ddmi})_{it} + \beta_4(\text{tdn_rump_ddmi})_{it} + \beta_5(\text{tdn_slobber_ddmi})_{it}$$

Where:

- **FCR** = Feed Conversion Ratio (feed intake per kg of weight gain)
- **hasBEF** = Indicator for beef-focused diet type
- **mw_ratio** = Metabolic weight ratio
- **tdn_silage_ddmi** = Total Digestible Nutrients from silage per daily dry matter intake
- **tdn_rump_ddmi** = Total Digestible Nutrients from grass per daily dry matter intake
- **tdn_slobber_ddmi** = Total Digestible Nutrients from concentrate per daily dry matter intake
- **day_diff_2** = Squared time interval effect

Weight Gain Model

$$\text{WG}_{it} = \alpha_i + \beta_1(\text{hasBEF_dmi})_{it} + \beta_2(\text{mw_ratio_dmi})_{it} + \beta_3(\text{tdn_silage_dmi})_{it} + \beta_4(\text{tdn_rump_dmi})_{it} + \beta_5(\text{tdn_slobber_dmi})_{it}$$

Where:

- **WG** = Total weight gain (**pred_weight_gain**)
- **hasBEF_dmi** = Indicator for beef-focused diet type, normalized by dry matter intake
- **mw_ratio_dmi** = Metabolic weight ratio normalized by dry matter intake
- **tdn_silage** = Total Digestible Nutrients from silage
- **tdn_rump** = Total Digestible Nutrients from grass
- **tdn_slobber** = Total Digestible Nutrients from concentrate
- **day_diff_2_dmi** = Squared time interval effect on dry matter intake

3.3 Model Derivation

Step 1: Initial Form

$$ADG = \frac{NE_g}{E_{req}} \quad (1)$$

Step 2: Expand Net Energy for Growth

$$ADG = \frac{ME - MEm}{E_{req}} \quad (2)$$

Step 3: Express ME in terms of Digestible Energy. Rewrite $ME = DE \cdot \alpha$ (where α is a coefficient), so during model training, we can use DE directly:

$$ADG = \frac{DE - MEm}{E_{req}} \quad (3)$$

Step 4: Substitute Gross Energy

$$ADG = \frac{GE - MEm}{E_{req}} \quad (4)$$

Step 5: Expand into Feed Components. Express gross energy as the sum of TDN (Total Digestible Nutrients) from different feed sources:

$$ADG = \frac{TDN_{slobb} \cdot \beta_1 + TDN_{grass} \cdot \beta_2 + TDN_{cornstalk} \cdot \beta_3 - MEm}{E_{req}} \quad (5)$$

Step 6: Express Maintenance Energy in terms of Metabolic Weight. Let $MEm = MW \cdot \beta_4$, where MW is metabolic weight and β_4 is the coefficient:

$$ADG = \frac{TDN_{slobb} \cdot \beta_1 + TDN_{grass} \cdot \beta_2 + TDN_{cornstalk} \cdot \beta_3 - MW \cdot \beta_4}{E_{req}} \quad (6)$$

Step 7: Linear Form for Statistical Modeling. For modeling purposes, we can rearrange to obtain a linear equation format:

$$ADG = \gamma_1 \cdot \left(\frac{TDN_{slobb}}{E_{req}} \right) + \gamma_2 \cdot \left(\frac{TDN_{grass}}{E_{req}} \right) + \gamma_3 \cdot \left(\frac{TDN_{cornstalk}}{E_{req}} \right) - \gamma_4 \cdot \left(\frac{MW}{E_{req}} \right) \quad (7)$$

step 8: get FCR and WG . To get those please not that WG can be interpreted as the time integral of ADG , while FCR can be interpreted as the derivative of feed intake with respect to weight gain (DMI derivative of ADG / WG), which allows derivation of the remaining interaction formulas. The interaction variables with time and DMI have been found by noting that:

$$ADG = \frac{WG}{Days}$$

$$FCR = \frac{WG}{DMI}$$

and so:

$$WG = FCR \times DMI$$

$$WG = ADG \times Days$$

Plugging in the values and regressing on the other target variables, making sure the system of variables holds under this structure, is how the interaction variables with time and DMI have been found.

3.4 3.3.2 Energy Required search

3.4.1 3.3.2.1 Equation for NEg required to gain 1 kg derivation

Step 1: Start with energy equation

$$\text{Energy (Mcal/d)} = a \times W^{0.75} + b \times W^{0.75} \times ADG^c$$

Step 2: Isolate the ADG term

$$b \times W^{0.75} \times ADG^c = -a \times W^{0.75} + \text{Energy (Mcal/d)}$$

$$ADG^c = \frac{-a \times W^{0.75} + \text{Energy (Mcal/d)}}{b \times W^{0.75}}$$

$$ADG^c = -\frac{a}{b} + \frac{\text{Energy (Mcal/d)}}{b \times W^{0.75}}$$

Step 3: Solve for ADG

$$ADG = \left(-\frac{a}{b} + \frac{\text{Energy (Mcal/d)}}{b \times W^{0.75}} \right)^{\frac{1}{c}}$$

Step 4: Define energy components

$$\text{Total Energy required (Mcal/d)} = a \times W^{0.75} + b \times W^{0.75} \times (ADG)^c$$

$$NEm = a \times W^{0.75}$$

$$NEg = \text{Total Energy} - NEm$$

Step 5: Define energy required for growth at a given ADG

$$\text{Total } NEg \text{ required at ADG (Mcal/d)} = b \times W^{0.75} \times \left(\frac{NEg}{NEgr} \right)^c$$

Step 6: Solve for NEgr

$$NEgr = b \times W^{0.75} \times \left(\frac{NEg}{NEgr} \right)^c$$

$$NEgr = b \times W^{0.75} \times \frac{NEg^c}{NEgr^c}$$

Step 7: Rearrange to isolate NEgr

$$1 = \frac{b \times W^{0.75} \times NEg^c}{NEgr^{(c-1)}}$$

$$NEgr^{(c-1)} = b \times W^{0.75} \times NEg^c$$

$$NEgr^{-1} = b \times W^{0.75} \times NEg^c$$

$$NEgr^{-1} = b \times (W^{0.75})^{1/c} \times NEg$$

Step 8: Linearization using logarithmic transformation

Taking the natural logarithm of both sides:

$$\ln(NEgr^{-1}) = \ln(b \times (W^{0.75})^{1/c} \times NEg)$$

$$-\ln(NEgr) = \ln(b) + \frac{0.75}{c} \times \ln(W) + \ln(NEg)$$

$$\ln(NEgr) = -\ln(b) - \frac{0.75}{c} \times \ln(W) - \ln(NEg)$$

Step 9: Express in linear form for regression

$$\ln(NEgr) = \beta_0 + \beta_1 \times \ln(MW) + \beta_2 \times \ln(NEg)$$

Where:

- $\beta_0 = -\ln(b)$
- $\beta_1 = -(0.75/c)$
- $\beta_2 = -1$

This is quite a complex feature, so first let's try to approximate it with:

3.4.2 3.3.2.2 Metabolic Weight

$$E_{req} = MW$$

Metabolic weight fails, it gives bad P-values, and in theory it would cause metabolic weight to disappear. Used because.

P and R values are worse.

Results can be verified by running:

```
python model_cli.py --model-type panel --model-name limousine_book_adg_1_final_MW --n
```

3.4.3 3.3.2.2 Root Of Dry Matter Intake

Assume the MW term is small, and the NEg is proportional to the root of DMI.

P and R values are worse.

Results can be verified by running:

```
python model_cli.py --model-type panel --model-name limousine_book_adg_1_final_root_d
```

3.4.4 3.3.2.3 Total Digestible Nutrients^{2/3}

Assume MW term is small and NEg term is proportional to $TDN^{2/3}$.

P and R values are worse.

Results can be verified by running:

```
python model_cli.py --model-type panel --model-name limousine_book_adg_1_final_2_3 --
```

3.4.5 3.3.2.4 NEg required approximately 1

$$NE_{gr} \approx 1$$

NEg required is approximately 1, this works.

Results can be verified by running:

```
python model_cli.py --model-type panel --model-name limousine_book_adg_1_final --n-we
```

3.4.6 3.3.2.5 Regressing against exponentials of ADG so the power c disappears

Makes heteroscedasticity worse. P and R values are worse.

Results can be verified by running:

```
python model_cli.py --model-type panel --model-name limousine_book_adg_1_final_2 --n-
```

3.4.7 3.3.2.6 Log transform

Log transform the whole formula so $\ln(NEgr)$ can be substituted. P and R values are worse.

Results can be verified by running:

```
python model_cli.py --model-type panel --model-name limousine_book_adg_1_final_log --
```

3.4.8 3.3.2.6 What could be tried and Conclusion

Hidden state models if more data is obtained.

However, assuming $NEgr \approx 1$ gives the best P-values, best R value, and is the simplest model.

3.4.9 3.3.2.3 Metabolic Weight $\times$ Metabolic Weight / Start Weight

Found this while trying to find E_{req} approximations.

My hypothesis for this is that the %fat increases as the ratio of metabolic weight / start weight increases, thus they are less heat resistant, which is a big factor in East Java.

Results can be verified by running:

```
python model_cli.py --model-type panel --model-name limousine_book_adg_1_final --n-we
```

3.5 Energy Required Search

Multiple failed attempts:

- $E_{req} = MW$
- $E_{req} = \sqrt{DMI}$
- $E_{req} = TDN^{2/3}$
- Exponential ADG regression
- Log-transformed system

Final working hypothesis:

$$NE_{gr} \approx 1$$

4 Results and Analysis

4.1 4.1 Model Performance Comparison

With Kalman filtering applied and the tails cut.

Model Type	Overall R^2	Within R^2	Cross-Val RMSE
Panel (Naive)	-27.832	0.5246	6.8415
Panel (Final ADG)	0.8762	0.6095	1.2271
Panel (Final FCR)	-0.1008	0.6435	8.7144
Panel (Final WG)	0.8660	0.5585	8.7144

Table 1: Model performance comparison after Kalman filtering and tail trimming

Model Type	Key Strength	Key Limitation
Panel (Naive)	Simplicity	Not based on scientific formula's
Panel (Final ADG)	Based on known formula's	Heteroscedasticity
Panel (Final FCR)	Based on known formula's	Heteroscedasticity
Panel (Final WG)	Based on known formula's	Heteroscedasticity

Table 2: Qualitative assessment of model strengths and limitations

4.2 Key Findings

Slight negative effect in the overall R^2 for FCR, indicating the cattle-specific effects dominate FCR.

High RMSE on the Cross Validation, indicating over-fitted data, and over-reliance on specific data points to make a good model or model misspecification.

4.2.1 Breed Effects

Only 2 Sementals were present in the dataset, so the model is not the most accurate; however, a simple model to reduce the chance of overfitting did confirm they perform significantly worse than Limousines.

4.2.2 Feed Component Analysis

Different feed components showed varying impacts on ADG:

- Grass-based TDN showed the highest efficiency ($\beta = 0.5625$)
- Silage and concentrate components showed lower but still positive effects
- Evidence of diminishing returns with increasing feed intake because of the negative DMI interaction effects

4.2.3 Weight and Growth Relationships

$$\text{mw_dmi_dt_ratio} \quad \beta = -0.0006$$

This means that the current metabolic weight, times the DMI it received, times the ratio between feedlot entrance weight and current weight, is negative.

4.3 4.3 Predictive Performance

Our final fixed effects model achieved:

- Within-sample R^2 of 0.712 (71.2% variance explained)
- Cross-validated RMSE of 1.265 kg/day is very high

This seems to be borderline unusable for new cattle predictions. because of the low sample size and the high differences between cattle. It seems to be really good for cattle predictions when the cattle has already has 2-3 measurements.

5 Discussion and Implications

5.1 5.1 Theoretical Implications

The results support established nutritional theories while providing quantitative insights:

- Confirmation of the metabolic weight scaling principle in cattle growth
- Evidence for breed-specific feed utilization pathways
- Quantify the effect of different feeds

5.2 5.2 Practical Applications

These findings have direct implications for feedlot management:

- **Economic Optimization:** Model coefficients enable calculation of profit-maximizing feeding strategies
- **Dry Matter Intake Optimization:** Ability to optimize DMI. Check: `dmi_optimization.py` tool
- **Optimum Exit Weight Optimization:** When the weight pressure pushes the ADG down so much that additional ADG is almost impossible

5.3 5.3 Limitations

- Limited breed diversity in the current dataset
- Potential seasonal effects not captured in the current model
- Limited data
- High cattle diversity within breed

6 Conclusion

The most effective model, a fixed effects panel approach focusing on ADG prediction, achieved strong predictive performance (Within $R^2 = 0.6095$, Overall $R^2 = 0.8762$) by incorporating biologically-derived variables based on established nutritional principles. The model reveals that ADG is significantly influenced by the metabolic weight to dry matter intake ratio, with feed component efficiency following a clear hierarchy: grass-based TDN ($\beta = 0.5625$) substantially outperforms both silage ($\beta = 0.3622$) and concentrate ($\beta = 0.2588$).

Individual animal effects account for approximately 26.6% of growth performance variance, highlighting the importance of genetic and developmental factors in cattle productivity. The substantial range in animal fixed effects demonstrates that cattle can differ by up to $\pm 50\%$ from average performance even under identical management conditions.

The negative between R^2 in the FCR model (-1.8908) confirms that while we can predict changes in feed efficiency within animals effectively (Within $R^2 = 0.6435$), predicting baseline efficiency between different animals remains challenging without additional genetic information.